Statistical Alignment of E8 Lattice Projections and Global Sacred Sites: A Monte Carlo Pipeline with Bonferroni Correction

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Abstract

This study investigates the spatial relationship between the E8 root system—an 8-dimensional polytope with 240 vertices and 6,720 edges—and a catalog of 160 globally distributed sacred sites. Using a rigorous "best-of-search" Monte Carlo pipeline that accounts for optimization bias, we identify four specific 8D-to-3D projections exhibiting statistically significant alignment with these sites ($p < 0.05$ after Bonferroni correction). The alignment is characterized as a fine-structure phenomenon residing in the precise angular relationship between individual E8 edges and site locations. Perturbation analysis, random-site controls, and cross-catalog validation confirm robustness. Although the signal meets conventional significance thresholds, it remains catalog-specific rather than a universal geometric law: different site arrangements resonate with different E8 projections. We discuss implications and outline a five-level research program for further investigation.

1 Introduction

The hypothesis that ancient and sacred sites follow higher-dimensional geometric principles has long been debated. The present study provides a rigorous statistical test using the E8 root system, a highly symmetric 8D lattice whose exceptional Lie algebra structure has attracted attention in both physics and geometry. The core research elements are:

- **The projection:** 8D vertices are projected into 3D via orthonormal matrices and mapped onto a unit sphere; edges become great-circle arcs.
- **The data:** A primary catalog of 160 sacred sites (temples, megaliths, pilgrimage centers) selected for cross-cultural significance without geometric filtering.
- **The metric:** Root-mean-square (RMS) distance from sites to the nearest projected E8 edge.

To avoid "search bias"—the tendency for any geometric network to align with points given sufficient orientation freedom—we employ a tournament pipeline with a best-of-search null model. Below we detail the methodology, confirmed projections, statistical robustness, and the fine-structure interpretation of the signal.

2 Methodology and Statistical Rigor

2.1 Best-of-Search Null Model

The critical innovation is a null model where each Monte Carlo trial generates a random site catalog (via SO(3) rotation) and then runs the entire optimization pipeline. This ensures that the null distribution reflects the best alignment achievable by pure chance under identical search parameters (latitude, longitude, bearing).

2.2 Tournament Pipeline Stages

1. **Coarse + refine search:** 270 seeds (projections) optimized over 23,328 candidate orientations.
2. **RMS gate:** Seeds must achieve RMS $\leq 0.0485°$ to proceed.
3. **Null testing:** 1,000-trial Monte Carlo tests against rotated site catalogs and quantile fraction analysis at five thresholds.
4. **Bonferroni correction:** Applied to all final p-values to account for multiple comparisons.

3 Key Findings: Confirmed Projections

Out of 270 seeds tested, four achieved definitive confirmation on the 160-site catalog. These seeds exhibit high effect sizes (Cohen's $d = 2.13$–3.52). Table 1 summarizes their performance.

Table 1: Confirmed seeds and their alignment signatures.

Seed	Orientation	Signature	Performance detail
3	65°N, 125°E, 250°	Breadth	96.9% of sites within 11 km (0.10°)
85	75°S, 75°W, 280°	Breadth	97.5% of sites within 11 km (0.10°)
48	5°S, 135°E, 250°	Precision	83.1% of sites within 5.6 km (0.05°)
46	55°N, 175°W, 250°	Ultra-precision	30.6% of sites within 1.1 km (0.01°)

3.1 Perturbation Stability

Confirmed seeds exhibit sharp, V-shaped RMS wells. A displacement of just 5° in any orientation parameter increases the RMS by 44–72%. In contrast, null optima (randomly generated) show near-zero sharpness (z-scores of 4.1–6.8). This demonstrates that the alignment is highly sensitive and specific to the optimized coordinates.

4 Analysis of the Alignment Signal

We systematically investigated why certain seeds succeeded while others failed.

4.1 What the Signal Is Not

- **Not matrix structure:** Random Forest and Logistic Regression on projection matrix features (row balance, entropy) gave AUC ≈ 0.52; matrix properties cannot predict success.
- **Not edge topology:** Aggregate metrics (edge length distribution, edge redundancy per site, angular coverage) were identical (within $< 3\%$ difference) between confirmed and failed seeds.
- **Not preferred bearings:** A chi-square test on bearings of shortlisted seeds confirmed uniformity ($p = 0.794$); no global orientation bias.

4.2 Fine-Structure Interpretation

The alignment is a fine-structure phenomenon: it arises from the precise interaction between specific E8 edges and specific site locations. The signal is visible only at the $0.01°$–$0.10°$ scale and is destroyed by minor orientation perturbations.

4.3 Dimension Fingerprinting

Successful seeds show a suggestive structural pattern:

- **Dimension 2:** Consistently strong (rank 1–3 in all confirmed seeds across both catalogs).
- **Dimension 4:** Consistently weak/suppressed (rank 7–8 in four of five confirmed seeds).

The combinatorial probability of this pattern recurring is $p = 0.0074$.

5 Cross-Catalog and Transfer Analysis

A secondary 62-site catalog (77% overlap with primary) was used to test universality.

- **Transfer failure:** The four confirmed 160-site seeds did not show significant alignment at fixed orientations against the 62-site catalog.

- **New success (Seed 89):** A fresh tournament on the 62-site catalog identified Seed 89 as uniquely exceptional (RMS $p = 0.002$, $d = 3.87$). Performance: 100% of sites within 11 km; median distance ≈ 2.1 km.

This confirms that alignment is projection-and-orientation-specific. Different site arrangements resonate with different subspaces of the E8 system.

6 Future Research Directions

We outline a five-level research program to isolate the mechanism behind these statistical observations:

1. **Level 1 (Independent Replication):** Test against geophysical features (tectonic plates, volcanic ridges) to see if the signal reflects Earth's underlying physical structure.
2. **Level 2 (Spatial Structure):** Analyze spherical harmonic decomposition of site catalogs to find shared spatial frequencies with E8 edge spacing.
3. **Level 3 (Synthetic Catalogs):** Test catalogs that preserve latitude density but randomize longitudes to isolate which spatial property E8 is detecting.
4. **Level 4 (Edge-Site Mapping):** Identify specific E8 vertex indices and edges involved in the alignment to find a "privileged substructure."
5. **Level 5 (Reverse Engineering):** Reconstruct the 8D direction the sacred-site signal points to by projecting only the "carrier" dimensions (e.g., Dimension 2).

7 Conclusion

The alignment of E8 lattice projections with sacred sites is a statistically robust, fine-tuned empirical observation. It survives the most rigorous search-bias corrections, including Bonferroni adjustment and perturbation analysis. While it defies current explanation and does not provide a universal physical mechanism, the study characterizes it as a specific resonance between higher-dimensional lattice topology and terrestrial site arrangements, rather than a generalized geometric law. The catalog-specific nature (exemplified by the transfer failure and the emergence of Seed 89) suggests that the phenomenon is not a universal constant but a conditional geometric relationship worthy of further investigation.

References

[1] Gherbi, S.-E. (2026). Statistically Significant Alignment Between E8 Lattice Projections and Sacred Site Locations on Earth (1.0.0). Zenodo. https://doi.org/10.5281/zenodo.19047661